CLASSICAL GENERATORS OF THE COMPACT OBJECTS VERSUS GENERATORS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11506-11527, following the statement at 11494-11504.

Editorial note. No omissions: the converse direction is fully argued through the two carded lemmas.

Allowed standard prerequisites (editorial).

- Tower construction from a generator. lemma-write-as-colimit (carded L26187, applied with I = {1}, cited at ~11510). Writes every compact X as hocolim of a tower built from E.
- Compactness and finite factorization. lemma-commutes-with-countable-sums (cited at ~11514) and lemma-factor-through (carded L26188, cited at ~11520). X factors through some X_n and then through a finite subfamily, i.e. X is in <E>.
- Right orthogonal criterion. lemma-right-orthogonal (same chapter, cited at ~11503). Gives (1) => (2) formally.

Source reference. Original lemma 094A: sources/stacks-derived/pinned-original-source.tex, source lines 10623–10623. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09SP: sources/stacks-derived/pinned-original-source.tex, source lines 11432–11432. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09SK: sources/stacks-derived/pinned-original-source.tex, source lines 11251–11251. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09SN: sources/stacks-derived/pinned-original-source.tex, source lines 11386–11386. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proposition 0.1. *Let $\mathcal{D}$ be a triangulated category with direct sums. Let E be a compact object of $\mathcal{D}$. The following are equivalent*

(1) *E is a classical generator for $\mathcal{D}_c$ and $\mathcal{D}$ is compactly generated, and*
(2) *E is a generator for $\mathcal{D}$.*

Proof. If E is a classical generator for $\mathcal{D}_c$, then $\mathcal{D}_c = \langle E \rangle$. It follows formally from the assumption that $\mathcal{D}$ is compactly generated and Lemma 09SK that E is a generator for $\mathcal{D}$.

The converse is more interesting. Assume that E is a generator for $\mathcal{D}$. Let X be a compact object of $\mathcal{D}$. Apply Lemma 09SN with $I = \{1\}$ and $E_1 = E$ to write

$$X = \text{hocolim} X_n$$

as in the lemma. Since X is compact we find that $X \to \text{hocolim} X_n$ factors through X_n for some n (Lemma 094A). Thus X is a direct summand of X_n. By Lemma 09SP we see that X is an object of $\langle E \rangle$ and the lemma is proven. □